

Vikram Anantha

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Education

University of Michigan	Aug. 2024 – May 2028 (expected)
<i>B.S. Computer Science & Electrical Engineering</i>	3.94/4.00 GPA
• Relevant coursework: Logic Design, Differential Equations, Machine Learning, Computer Vision, Data Structures & Algorithms, Digital Circuits, Computer Organization, Linear Algebra, Discrete Math • Clubs: Head Photo Editor of the Michiganensian Yearbook, Active member of Tau Beta Pi Engineering Honors Society • Honors: 1 of 6 recipients of 1931e Merit Scholarship for Engineering, James B. Angell Scholar for 3 consecutive terms	

Industry Experience

Meta	May. 2026 – Jul. 2026
<i>Software Eng. Intern</i>	Menlo Park, CA
• Architected Gaussian Splatting pipeline for 3D models of UofM's Mcity, cutting AV training data costs • Built hierarchical Gaussian Splat map visualization, managing a team of 3 student engineers	
Capoom	Jan. 2025 – Dec. 2025
<i>Software Eng. Intern</i>	Ann Arbor, MI
• Architected Gaussian Splatting pipeline for 3D models of UofM's Mcity, cutting AV training data costs • Built hierarchical Gaussian Splat map visualization, managing a team of 3 student engineers	
Stellantis	May 2025 – Aug. 2025
<i>ADAS Functional Integration Eng. Intern</i>	Auburn Hills, MI
• Deployed CAPL script to analyze CAN-bus signals, identifying a 10% fault rate for ADAS teams • Designed MATLAB Simulink model to control CAN-bus signals, quantifying self-driving behavior	
Constant Therapy Health	Jun. 2024 – Aug. 2024
<i>Software Eng. Intern</i>	Newton, MA
• Fine-tuned Wav2Vec ASR on aphasia phonemes, achieving 4x lower false negative rate than Google's • Built task visualizer and LLM grammar detector to support stroke patient speech therapy	

Research Experience

University of Michigan, ROAHM Robotics Lab	Aug. 2024 – Present
• Built real-time 3D segmentation (Gaussian Splat, SAM-2, CLIP) via ROS2 for safe robotic arm use • Developing Transformer-based VLA models for generalist robot policies across diverse tasks	
Columbia University, Center for Smart Streetscapes	May 2023 – Feb. 2024
• Developed CV + MQTT app to warn pedestrians of collisions; co-authored 2024 IEEE IV Symposium paper • Optimized alert pipeline to <1.5s latency, meeting human reaction time to prevent accidents	
Massachusetts Institute of Technology, Auto ID Lab	Jun. 2023 – Sep. 2023
• Built YOLOv8-based 2FA system to verify AV headlight patterns, securing V2I communication	

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, CAPL, ReactJS, Assembly, Bash, LaTeX
Frameworks & Tools: Robot Operating System (ROS & ROS2), Amazon Web Services (AWS), Vector CANalyzer, MATLAB/Simulink, OpenCV, TensorFlow/PyTorch, Git/GitHub, CUDA, Docker, Gaussian Splatting, LLMs, YOLOv8 Object Detection, SAM-2 Segmentation, SLAM

Selected Honors & Awards

Immerse The Bay Hackathon, Stanford University, 1st Place Winner	Nov. 2025
American Invitational Mathematics Exam (AIME), 2 Time Qualifier	Mar. 2021 & Feb. 2022
USA Coding Olympiad (USACO), Silver Division	Dec. 2019

Publications

Digital Twin for Pedestrian Safety Warning at a Single Urban Traffic Intersection	2024
• V. Anantha, et al., 2024 IEEE Intelligent Vehicles Symposium (IV), pp. 2640-2645, doi: 10.1109/IV55156.2024.10588544	
Peer to Peer Learning Platform Optimized With Machine Learning	2022
• V. Anantha, arXiv:2209.03489, 2022	